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Does Bitcoin lead to Diversification Gains? Evidence from India

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Abstract

Achievement of diversification gains is one of the most important objectives of portfolio optimization. In this study, we aim to find out the extent and significance of diversification gains that accrue to an already diversified portfolio due to inclusion of Bitcoin. We use global indices to build a portfolio spread across six asset classes namely money market, equity, fixed income, commodities, real estate and alternative investments for adequate representation of existing investment choices. To construct optimal portfolios, we employ three investment strategies viz. long only, -100%/+100% & equally weighted. Instead of the traditional mean-variance optimization methodology, we apply Mean-CVaR framework for optimal allocation of weights to assets. Our findings suggest that portfolios containing Bitcoin have superior risk-adjusted returns as compared to portfolios without Bitcoin for long only and equally weighted strategies. The consistency of higher risk-return ratio across time in these strategies indicates robust diversification benefits of the crypto-currency for a portfolio denominated in Indian rupee. The study is relevant for individual and institutional investors as it highlights the importance of inclusion of Bitcoin in optimal portfolios. The study also adds fresh evidence to the scarce literature on Bitcoin from an investment perspective.

Key Words: Bitcoin, Portfolio Optimization, Expected Shortfall, Modified Sharpe Ratio

JEL Classification Codes: G11, G15, C61

1. Introduction

The quest for finding new assets that provide diversification gains has intrigued academicians and practitioners since (Markowitz, 1952) introduced the idea of an ‘efficient portfolio’. A recent development in this regard is the emergence of a peculiar asset, Bitcoin. Although (Nakamoto, 2008) floated the concept of a decentralised virtual currency with an intention to challenge the loopholes of the current monetary system, Bitcoin has also grabbed headlines for its phenomenal returns and low correlation with other traditional asset classes. As on 30th June 2017, it has delivered a staggering return of 149% in past six months and 260% in past one year¹ for investments denominated in INR. There is an increasing interest for world’s most popular virtual currency among tech entrepreneurs as Bitcoin startups in India including Zebpay, Coinsecure and Unocoin have raised investments in excess of one million dollars each in 2016². In its Financial Stability Report (December 2015), the RBI recognised the importance of the blockchain technology to disrupt financial markets, collateral identification and payment systems. The bank asserted the need for regulators and authorities to keep pace with the developments in the blockchain technology as it possesses the potential to transform back office operations of banks and improve the speed and cost efficiency in payment systems and trade finance.

2. Literature Review

Being one of first studies to evaluate the diversification benefits of Bitcoin, (Brière, Oosterlinck, & Szafarz, 2015) show that inclusion of even a small proportion of the virtual currency may dramatically improve the risk-return characteristics of a portfolio. Using data over the period

¹ Data for Bitcoin price in USD taken from <https://www.coindesk.com> and USD/INR exchange rate data taken from Bloomberg

² <https://inc42.com/buzz/bitcoin-cryptocurrency-india-government/>

between 2010 and 2013, they report higher Sharpe and Sortino ratios on inclusion of Bitcoin from the standpoint of a US investor. They use mean-variance spanning test suggested by (Ferson, Foerster & Keim, 1993) and (Huberman and Kandel, 1997) to gauge diversification benefits of addition of Bitcoin in a diversified portfolio. They show BTC-inclusive mean-variance efficient frontier to be much steeper than its BTC-free counterpart.

(Eisl, Gasser, & Weinmayer, 2015) test the relevance of addition of Bitcoin to an already well diversified portfolio comprising of 12 indices to represent different asset equity, fixed income, money market, commodities, real estate and alternative investments. They follow (Rockafellar, 1997) and use Conditional Value at Risk as the proxy for risk keeping in view the non-normal distribution of Bitcoin returns. With a perspective of an US investor, they calculate risk-return ratio similar to (Campbell, Huisman, & Koedijk, 2001) to find the optimal combination of assets on a rolling basis for each month from July 2010 to April 2015. They conclude that average monthly returns and CVaRs are higher for a portfolio with Bitcoin. With average weights for Bitcoin ranging from 1.65% to 7.65% across different portfolio construction methods, the study reports superior risk-adjusted performance of portfolios containing Bitcoin. (Chowdhury, 2016) assesses Bitcoin's role in financial markets. As one of the initial attempts to evaluate inclusion of Bitcoin in a portfolio, it follows a general approach than the traditional mean-variance in-sample setting. They have followed the approach suggested by (Daskalaki & Skiadopoulos, 2011) and employed rigorous spanning. Results suggest Bitcoin offers diversification benefits to investors regardless of the performance measure used.

(Moore & Stephen, 2016) assess the potential benefits and costs of holding Bitcoins as a part of portfolio of international reserves by considering the case of Barbados. They have applied two main empirical tools to execute the analysis: a counterfactual exercise using historical

performance of the various exchange rates (including Bitcoin) and a Monte Carlo forecast of international reserves for the next 1, 2, 5 and 10 years which consists of 0.01% Bitcoin portfolio composition. The study found that if Bitcoin is included in the portfolio it would have generated significant returns but the volatility of the reserves would also increase. Therefore, the share of Bitcoin must be small in the portfolio of foreign balances for the Central Bank of Barbados. (Rajan & Rana, 2011) studied and implemented the robust method for portfolio optimization in the Indian stock market. They have followed empirical analysis to generate an optimal portfolio for an investor as well as followed (Ledoit & Wolf) to generate test performances as “rolling horizon method”. The daily historical log returns data for July 2001 to June 2011 period. CAPM model is used for in-sample and out-of-sample period rather than taking the average of returns. The findings show that an investor can make reasonable profit by construction of a fund that consists of a fixed income asset and with around 14 stocks. The analysis is performed only on without short selling case.

3. Research Problem and Research Gap

Most of the existing academic literature has focussed on the legal and information technology aspects of Bitcoin. In this study, we attempt to ascertain the diversification gains that accrue to an already well diversified portfolio from the perspective of an Indian investor. Considering the thin body of literature on Bitcoin from a financial economics point of view, we believe the paper adds value to our understanding on the benefits of investments in the virtual currency.

4. Objective

Considering the exceptional growth in value that Bitcoin has witnessed (specifically since August 2015), investor community has shown immense interest in the currency as a fresh avenue for asset allocation. For an investor holding a diversified portfolio of assets, inclusion of Bitcoin depends on the nature and extent of its impact on the return-risk ratio of the portfolio. Hence, we aim to answer the following research questions in this study.

1. Does inclusion of Bitcoin improve the return-risk ratio of an already well diversified portfolio?
2. Is Bitcoin's weight in an already well diversified portfolio robust to the investment strategy used?
3. How does Bitcoin's weight in an already well diversified portfolio behave over time?

We approach these above-mentioned issues from the perspective of an Indian investor. This implies the measurement of index proxies of all asset classes as well as Bitcoin in INR.

5. Data and Research Methodology

The historical Bitcoin price data becomes available from July 18, 2010 on CoinDesk which reports a simple average of global Bitcoin/USD exchange prices and hence our sample period covers 77 months until December 30, 2016. To ensure adequate representation of each prominent asset class, we choose at least one leading liquid index as a proxy for the same. Data for all indices have been taken from Bloomberg. All assets have been measured in INR and USD/INR exchange rate has also been taken from Bloomberg. The list of assets considered to make up a well-diversified portfolio is provided in Table 1.

S. No.	Asset Class	Indices
1.	Equity	MSCI World Index
2.	Fixed Income	JPMorgan Global Government Bond Index
3.	Fixed Income	Bloomberg Global Investment Grade Corporate Bond Index
4.	Real Estate	MSCI World Real Estate Index
5.	Alternative Investments	Morningstar Diversified Alternatives Index
6.	Alternative Investments	Global Hedge Fund Index
7.	Commodities	Thomson Reuters Core Commodity CRB Index
8.	Money Market	Deutsche Bank Currency Returns Index
9.	-	CoinDesk Bitcoin Price Index

Table 1: The table lists the assets considered in the study to construct a diversified portfolio. Data for all indices has been taken from Bloomberg database. Data for Bitcoin has been taken from CoinDesk Bitcoin Price Index. All values are quoted in INR in the study. The sample period starts on July 18, 2010 and ends on December 31, 2016.

Keeping in view the non-normal distribution of Bitcoin returns, we follow (Rockafellar, 1997) and use Conditional Value at Risk as the proxy for risk. We follow (Eisl, Gasser, & Weinmayer, 2015) to construct optimal portfolios based on the asset returns of the previous one year. Hence, for the period July 2010 to July 2011, we do not calculate any weights. Keeping our rolling window size as 12 months, we calculate optimal portfolio weights for a total of 66 months as we move the window forward by one month each time. Using the optimal weights obtained by mean-CVaR optimisation for a particular month, we calculate out-of-sample monthly return and CVaR. We perform this procedure for three portfolio construction strategies namely long only, -100%/+100% and equally weighted. For each strategy, we form two portfolios: one with Bitcoin and other without Bitcoin. In this way, we calculate 66 return-risk ratios similar to (Campbell et al., 2001) for each investment strategy i.e. a total of eight different combinations (two for each strategy). We then make a comparison of the return-risk ratios of portfolios with and without Bitcoin for each investment strategy to ascertain if Bitcoin offers any diversification benefits.

6. Results and Findings

As represented in Figure 1, the price of Bitcoin remained broadly stable for the initial three years. The first notable spike occurred around October 2013 followed by a gradual decline through the first quarter of 2015. The Bitcoin growth story started towards the end of 2015 and picked up unprecedented pace in August 2016. In fact, Bitcoin has given a superlative return of 321% in the 10 month period from August 2016 to June 2017. This period also includes sharp spikes in price due to shocks such as the cyber ransom-ware attack in May 2017.

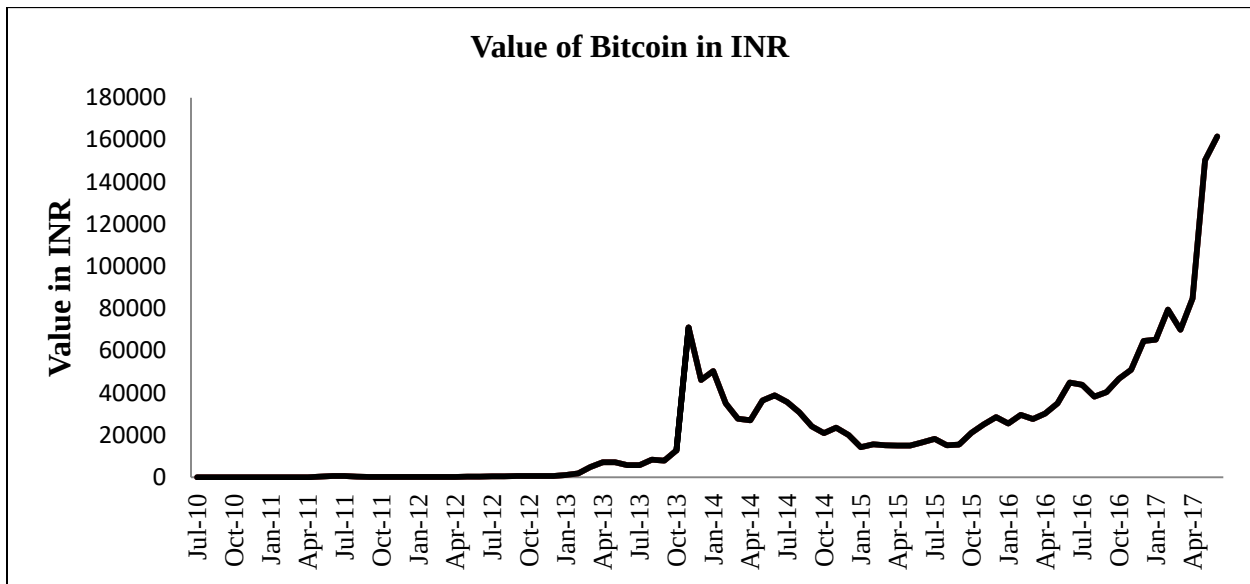


Figure 1: The graph shows Bitcoin’s price development in INR since its inception in July 2010 till June 2017. Data for prices of Bitcoin has been gathered from CoinDesk Bitcoin price index and USD/INR exchange rate values have been taken from Bloomberg.

In Table 2, we describe the empirical results of our study. For each of the three portfolio optimization techniques, we compare mean monthly return, mean monthly CVaR and mean monthly return-risk ratio for a portfolio with Bitcoin and another portfolio without Bitcoin. For all the three optimization techniques, the mean monthly return and mean monthly CVaR of the portfolio including Bitcoin is greater than the portfolio without Bitcoin. This implies that Bitcoin offers an upside potential to the portfolio return but at the cost of additional risk. To evaluate if the portfolio with Bitcoin is better in terms of a risk-return trade-off, we compare the return-risk

ratios of portfolio with and without Bitcoin. For long only and equally-weighted strategies, the portfolio with Bitcoin has a higher return-risk ratio. This indicates superior risk-adjusted returns for portfolios with Bitcoin against those without it. However, for -100%/100% strategy, the portfolio without Bitcoin offers better risk-adjusted returns as compared to the portfolio with Bitcoin. In summary, Bitcoin offers diversification gains in two out of three portfolio optimization techniques considered in the study.

Portfolio Optimisation Framework	Mean Monthly Bitcoin Weight	Mean Monthly Return	Mean Monthly CVaR	Mean Monthly Return-Risk Ratio
Long Only With Bitcoin	9.89%	5.47%	2.00%	4.35
Long-Only Without Bitcoin	-	2.42%	2.02%	1.56
-100%/100% With Bitcoin	3.00%	5.30%	1.98%	10.12
-100%/100% Without Bitcoin	-	4.98%	1.31%	11.07
Equally weighted With Bitcoin	11.11%	5.00%	12.65%	1.21
Equally Weighted Without Bitcoin	-	1.70%	3.46%	0.5276

Table 2: The table depicts final empirical results of the study in the form of mean monthly return, mean monthly CVaR, mean monthly return-risk ratio and mean monthly Bitcoin weights for all three optimization techniques considered in the study

With regard to the mean monthly weight of Bitcoin in the three strategies, we report an average weight of 9.89% and 3.00% for long only and -100%/100% strategy respectively. For equally weighted portfolio, we simply get a constant weight of 11.11% as there are nine assets including Bitcoin that form part of the analysis. This indicates a wide variation in Bitcoin weights across the three strategies. Hence, an investor who does not consistently follow a single strategy may have to rebalance his portfolio frequently and this in turn may lead to higher transaction costs.

Our results suggest that it is prudent for investors to include Bitcoin in their portfolio if their investment strategy is either long only or equally weighted as portfolios with Bitcoin deliver higher risk-adjusted returns for these two strategies. However, investors who follow -100%/+100% strategy should not include Bitcoin in their already diversified portfolio as inclusion of Bitcoin leads to a more proportionate increase in risk than return.

7. Conclusion and Implications

In this study, we aim to find out the extent and significance of diversification gains that accrue to an already diversified portfolio due to inclusion of Bitcoin. We use global indices to build a portfolio spread across six asset classes namely money market, equity, fixed income, commodities, real estate and alternative investments for adequate representation of existing investment choices. To construct optimal portfolios, we employ three investment strategies viz. long only, -100%/+100% & equally weighted. Instead of the traditional mean-variance optimization methodology, we apply Mean-CVaR framework for optimal allocation of weights to assets. Our findings suggest that portfolios containing Bitcoin have superior risk-adjusted returns as compared to portfolios without Bitcoin for long only and equally weighted strategies. The consistency of higher risk-return ratio across time in these strategies indicates robust diversification benefits of the crypto-currency for a portfolio denominated in Indian rupee. The study is relevant for individual and institutional investors as it highlights the importance of inclusion of Bitcoin in optimal portfolios. The study also adds fresh evidence to the scarce literature on Bitcoin from an investment perspective.

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